



Intergenerational Transmission of Social Anxiety: The Role of Parents' Fear of Negative Child Evaluation and Their Self-Referent and Child-Referent Interpretation Biases

Melis Dülger^{1,2} · Bram Van Bockstaele^{1,3,4} · Mirjana Majdandžić¹ · Wieke de Vente¹

Accepted: 9 April 2024
© The Author(s) 2024

Abstract

Purpose Parents' fear of negative evaluation (FNE), fear of negative child evaluation (FNCE; parents' fear that their child is evaluated negatively by others), and self- and child-referent negative interpretation biases have been proposed to play a role in the intergenerational transmission of social anxiety. In this cross-sectional study, we tested whether parents' self-reported FNE and self-referent interpretation bias, as well as their FNCE and child-referent interpretation bias, statistically mediated the relationship between parent social anxiety and parent-reported child social anxiety.

Methods A total of 179 parents of 13–16-year-old adolescents completed questionnaires concerning own social anxiety, their FNE, and FNCE and their child's social anxiety. Parents' self- and child-referent interpretation biases were measured using scenario completion and memory recognition tasks.

Results Parents' FNCE partially statistically mediated the relationship between parent and child social anxiety. Parents' FNE and their self- and child-referent interpretation biases did, however, not statistically mediate this relationship.

Conclusions Parents' FNCE does, but their FNE and self- and child-referent interpretation biases do not seem to play a role in the intergenerational transmission of social anxiety. Hence, parents' FNCE might be a promising target for clinical practice while designing therapies and interventions concerning child social anxiety.

Keywords Fear of Negative Child Evaluation · Fear of Negative Evaluation · Social Anxiety · Interpretation bias · Parents

✉ Melis Dülger
M.Dulger@uu.nl

Bram Van Bockstaele
Bram.VanBockstaele@UGent.be

Mirjana Majdandžić
M.Majdandzic@uva.nl

Wieke de Vente
W.deVente@uva.nl

¹ Research Institute of Child Development and Education, University of Amsterdam, Nieuwe Achtergracht 127, Amsterdam 1018 WS, The Netherlands

² Present address: Department of Education, Utrecht University, Heidelberglaan 1, Utrecht 3584 CS, The Netherlands

³ Centre for the Advancement of Research on Emotion, School of Psychology, University of Western Australia, 35 Stirling Highway, Crawley, WA 6009, Australia

⁴ Department of Developmental, Personality, and Social Psychology, Ghent University, Henri Dunantlaan 2, Ghent 9000, Belgium

Socially anxious individuals tend to avoid or feel discomfort during social interactions because they are concerned that they will say or do something humiliating or embarrassing (Stein & Stein, 2008). This fear of negative evaluation (FNE) is widely acknowledged as one of the core features of social anxiety, which is reflected in the diagnostic criteria of social anxiety disorder in the DSM-5-TR (American Psychiatric Association, 2013). With a lifetime prevalence of 8.6% in adolescents aged 13–17 in the US, social anxiety disorder is one of the most prevalent psychological disorders (Kessler et al., 2012) and can have serious long-term adverse effects on individuals' social and academic functioning (Bögels et al., 2011; de Lijster et al., 2018). The worldwide prevalence of social anxiety disorder ranges from 0.5 to 7% (American Psychiatric Association, 2013).

Research on anxiety shows that it tends to run in families (Murray et al., 2008) and is widespread among adolescents (de Lijster et al., 2018). Previous studies indicate

that parent and child social anxiety are strongly associated (e.g., Lieb et al., 2000; Telman et al., 2018). Besides genetic factors (Scaini et al., 2014), studies indicate that other parental influences contribute to the development and maintenance of social anxiety in young children (e.g., Ahmadzadeh et al., 2019; Eley et al., 2015). One factor that may be linking parental to child social anxiety is parents' fear of negative child evaluation (FNCE; Schreier & Heinrichs, 2010). FNCE refers to parents' fear that their child will be judged and evaluated negatively by others. Schreier and Heinrichs (2010) reasoned that socially anxious parents' FNE would extend to their children's environment, resulting in FNCE.

As socially anxious individuals typically expect others to be critical about themselves, they may also expect others to be critical about their children (de Vente et al., 2011). Therefore, parents' FNE might extend to their child's environment. Indeed, a recent cross-sectional study conducted by de Vente et al. (2022) with a sample of families with 7.5-year-old children showed that parents' FNE mediated the relation between parent social anxiety and FNCE for both mothers and fathers, supporting the notion that parents with high FNE are not only afraid of being negatively evaluated by others, but also afraid of their child being negatively evaluated by others. Both parents' FNE and FNCE may influence their child social anxiety through a (non)verbal information pathway (e.g., providing threat information). Indeed, studies on anxiety demonstrated that parents' own beliefs show similarities with children's beliefs (Bögels et al., 2003; Creswell et al., 2006; McDowell et al., 2002). De Vente et al. (2022) also found that fathers' (but not mothers') social anxiety was associated with their child's social anxiety via fathers' FNCE. This study provided support for the role of parents' FNCE in intergenerational transmission of social anxiety and for the assumption of parents' FNE extending to the child's environment, resulting in FNCE. Although parents' FNE may influence the transmission of social anxiety through FNCE, as it has already been shown by the study of de Vente et al. (2022), the lack of full mediation leaves room for other pathways than through FNCE. Parents' FNE, but not their FNCE, may also affect their child's social anxiety through modelling of anxious behaviors. This nonverbal transmission may occur through observing parents' reactions to ambiguous situations (Aktar et al., 2017). Yet, none of the previous studies addressed the individual role of parents' FNE in the association between parent and child social anxiety. Likewise, FNCE, but not FNE, may also affect child social anxiety through parents' verbal and nonverbal signals (e.g. encouraging versus anxious) in situations where the parent is present, while a child is handling a social interaction, as explained by Schreier and Heinrichs (2010). Hence, we expect the pathways through

FNE and FNCE to operate at least partially independently from one another.

With regard to the role of FNCE in the intergenerational transmission of social anxiety, Schreier and Heinrichs indeed found that mothers' (but not fathers') FNCE mediated the relation between maternal social anxiety and 9- to 16-year-old children's social anxiety in their cross-sectional study. This finding endorses the idea that FNCE at least partly explains the intergenerational transmission of social anxiety. In another study (based on data from the same longitudinal study as de Vente et al., 2022), de Vente et al. (2011) longitudinally investigated the mediating role of parents' FNCE on the association between parents' social anxiety and later infant social fear in families with their 1-year-old child. Parental social anxiety before their infant was born predicted parents' FNCE, confirming that socially anxious parents also worry more about others' negative opinions regarding their children. Crucially, fathers' FNCE partially mediated the relation between fathers' social anxiety and infant social fear. Even though the same effect was not found for mothers, these findings further support the notion that parents' FNCE at least partly explains the relation between parental and child social anxiety.

While FNE and FNCE concern fear beliefs that operate more explicitly and can be assessed through self-report, theories and research of the past few decades have also stressed the role of more automatic and implicitly measured processes in the development and maintenance of psychopathology, such as interpretation bias (for a review, see Roefs et al., 2011). Conceptually, FNE and FNCE are fear beliefs concerning perceived threat of social judgement, criticism, or rejection that reflect cognitions that are typically more strongly present in people with higher levels of social anxiety (Schreier & Heinrichs, 2010), whereas interpretation bias is a reflection of cognitive tendencies linked to how people perceive and interpret ambiguous social cues or events (Beck et al., 2005). Dual process models (e.g., Strack & Deutsch, 2004) suggest that these implicit and explicit cognitive processes operate in parallel with each other, and both could potentially explain different aspects of psychopathology. While implicit processes are fast, demand minimal cognitive effort, have a lower threshold for information processing, and are typically measured indirectly (e.g., using reaction times), explicit processes are slower, demand more cognitive effort, have a higher threshold, and are typically measured with self-reports (Strack & Deutsch, 2004; van Bockstaele et al., 2021). Occasionally, implicit and explicit processes may align closely and exhibit a strong connection, whereas in other instances, they may deviate and exist independently. Both implicit and explicit processes are considered valid indicators that can provide distinct insights into social anxiety (Teachman & Allen, 2007). Cognitive models

have identified several implicit biases in information processing characterizing anxiety in general (Williams et al., 1997) and social anxiety specifically (e.g., Clark & Wells, 1995; Heimberg et al., 2010). Compared to non-socially anxious individuals, socially anxious individuals are more likely to interpret ambiguous social cues as negative or threatening (for a review, see Hirsch et al., 2016). A recent systematic review and meta-analysis revealed a large effect size for the association between this negative interpretation bias and social anxiety (Chen et al., 2020), demonstrating that socially anxious individuals do indeed interpret ambiguous social information more negatively.

Interpretation bias has been studied using two types of tasks, measuring so-called *online* and *offline* interpretation processes (e.g., Chen et al., 2020). *Online* interpretation processes are typically operationalized as response latencies to stimuli, which makes this type of measurement less subject to demand, selection, or response bias effects. *Offline* interpretation processes are usually measured using self-reported interpretations of specific situations, either with or without retrieving information from one's memory about the particular situation. The advantage of *offline* interpretation processes is that participants get time to judge and reflect on the situation, facilitating more thorough information processing. Chen et al. (2020) found that studies using *offline* measures yielded higher effect sizes between social anxiety and interpretation bias than studies using *online* measures. Thus, the different ways of assessing interpretation bias may explain the heterogeneity in results in previous studies.

Research has shown that anxious parents' negatively biased interpretations are not limited to their own environment but extend to their children's environment. Lester et al. (2009) proposed that parents with high levels of trait anxiety may be inclined to interpret ambiguous situations as more threatening, meaning that they may have an interpretation bias. This tendency could potentially influence their children, leading them to develop a similar tendency towards interpreting ambiguous information as more negative. In their cross-sectional study, they examined whether anxious parents had interpretation biases concerning both their own and their children's environment. They found that higher levels of parent trait anxiety were related to more self-referent (i.e., the situation concerns the parent) and child-referent (i.e., the situation concerns the child) negative interpretation bias. Their study also showed a mediation effect of parents' self-referent interpretation bias in the association between parent trait anxiety and their child-referent interpretation bias. This supports the notion that anxious parents' interpretation bias also extends towards their children's environment. Although Lester et al.'s (2009) study indicated an extension of self-referent interpretation bias towards child-referent interpretation

bias, it focused on trait anxiety, and it did not investigate the role of parents' interpretation biases in the context of intergenerational transmission of *social* anxiety. We propose that a similar transmission may also occur in families with high social anxiety.

A recent meta-analysis of Subar and Rozenman (2021) involving 8 studies with samples of 4–17-year-old children and their parents indicated a small and significant correlation between parents' self-referent interpretation bias and child social anxiety, which confirms the relation between parents' interpretation bias and child social anxiety. One of the cross-sectional studies included in this meta-analysis, conducted by Podina et al. (2013), investigated mothers' self-referent negative interpretation bias as a *mediator* in the association between mothers' social anxiety and self-reported anxious symptoms in children, in middle childhood and early adolescence ages. They found a partial mediation effect for mothers' self-referent interpretation bias, supporting the idea that parents' negative interpretation bias links parent social anxiety to child social anxiety. However, this study and previous studies did not investigate the role of parents' *child*-referent interpretation bias in the intergenerational transmission of social anxiety. In addition, most of the studies involved in the meta-analysis investigated the association between mothers' interpretation bias and their child's social anxiety (Subar & Rozenman, 2021). It may be vital to also consider fathers' interpretation biases to further examine the role of interpretation bias in the development of child social anxiety.

Considering the evidence that parents' interpretation biases may extend to their child's environment, it might be crucial to investigate the role of parents' child-referent interpretation biases in explaining the relations between parent and child social anxiety. In addition, although previous studies investigated the roles of parents' FNE, FNCE, and interpretation bias separately in the social anxiety context, none of the studies investigated their roles in parallel. Thus, in the current study, we add to the knowledge about the intergenerational transmission of social anxiety by examining the role of parents' explicit FNE and FNCE, and their implicit self- and child-referent interpretation biases in parallel. Parents' fears and their interpretation biases could influence their child's social anxiety through the transfer of verbal and nonverbal threat information (Aktar et al., 2017; Field, 2006; Field et al., 2007). However, as parents may have different explicit and implicit beliefs and interpretation tendencies regarding their own and their child's environment, their reactions may vary in situations involving themselves versus their children. In addition, while the path through parents' FNE depends more strongly on

parents' own characteristics, the path through FNCE is presumably shaped more strongly by the child characteristics, particularly their social competence. For instance, a child with higher social competency is likely to trigger less FNCE in parents. A similar pathway can also apply parents' self-referent and child-referent interpretation biases. While parents' self-referent interpretation bias is presumably influenced by their own characteristics, parents' child-referent interpretation bias may be shaped more strongly by child characteristics. More specifically, we suggest two pathways that could potentially explain the relation between parent and child social anxiety: (a) parents' FNE and their *self*-referent interpretation bias may at least partially mediate (Podina et al., 2013) the association between parental and child social anxiety; and (b) parents' FNCE and their *child*-referent interpretation bias may at least partially mediate (see de Vente et al., 2011; Schreier & Heinrichs, 2010) the relation between parental and child social anxiety. While the first pathway concerns mediating effects of parents' explicit FNE and implicit self-referent interpretation tendencies, the second pathway concerns the mediating effects of parents' explicit FNCE and implicit child-referent interpretation tendencies.

Our main aim was to address the potential roles of parents' FNE and FNCE as well as self-referent and child-referent negative interpretation biases in explaining the intergenerational transmission of social anxiety. We focused on parents of adolescents because social anxiety increases during adolescence and the onset of social anxiety disorder typically peaks by mid-adolescence (Stein & Stein, 2008). Parents of adolescents completed self-report measures of social anxiety, FNE, FNCE, and their child's social anxiety. Parents also completed online and offline interpretation bias tasks measuring both their self-referent and child-referent interpretation biases. Because of the various strengths and weaknesses of both types of measures, we decided to use both of them to measure interpretation bias in this study. We expected positive relations between all constructs, with socially anxious parents having more socially anxious children and showing increased FNE, FNCE, self-referent, and child-referent negative interpretation biases. We tested the roles of these explicit and implicit processes in parallel using path models. We hypothesized that parents with higher social anxiety would have higher FNE and stronger self-referent negative interpretation biases, which in turn would increase their child's social anxiety. Similarly, we expected that parents with higher levels of social anxiety would have higher FNCE and stronger child-referent negative interpretation biases, which in turn would increase their child's social anxiety.

Method

Participants

A total of 182 parents (57.54% mothers; 41.90% fathers; 0.56% non-binary) of adolescents between 13 and 16 years were recruited through Prolific, an online data collection and participant recruitment platform. An a priori power analysis was conducted using Monte Carlo power analysis simulation (Schoemann et al., 2017) to determine an adequate sample size. With a significance criterion of $\alpha=0.05$, a sample size of 200, and assuming moderate correlations (around 0.30) between variables, an adequate power of around 0.80 was achieved for mediation models. The recruitment process was ended at 182 participants due to time constraints. The screening tool of Prolific was used to select eligible participants. In addition, demographic questions regarding child age, English proficiency level, and time spent with the child per week were included to eliminate participants who did not fit the eligibility criteria. The mean age of their children was 14.03 years ($SD=0.96$). In our sample, 44.69% of the parents had a daughter, and 54.75% had a son (0.56% non-binary). Only parents spending at least two days per week with their children and with adequate English language skills were included in the study. The majority of the parents (86.03%) lived in the same household with their child. Most parents had British (39.66%), American (20.11%), or South African (12.29%) backgrounds and identified as Caucasian (80.45%).

Procedure

Parents who met the inclusion criteria were informed of and approved the subject of the current study and general procedure through a digitally signed informed consent form. The study was conducted in the online platform Inquisit. After providing demographic information, the study started with the *online* (i.e., scenario completion) and *offline* (i.e., memory recognition) interpretation bias tasks, with a 2-minute break in between. Next, parents completed questionnaires on FNCE, FNE, child social anxiety, and own social anxiety. Finally, participants were debriefed and compensated with £6 for completing the study. The entire procedure was approved by the ethics committee of the University of Amsterdam (ref no. 2020-CDE-12913).

Materials

Child Social Anxiety

Child social anxiety was measured using parent report via the social anxiety subscale of the Screen for Child Anxiety

Related Emotional Disorders (SCARED; Birmaher et al., 1997). This subscale consists of 9 items and parents rate their child's social anxiety symptoms on a 3-point Likert scale (ranging from 1 = "almost never" to 3 = "often"). Child social anxiety was calculated by taking the sum scores of questionnaire items, with higher scores indicating higher child social anxiety. An example item is: "My child feels nervous with people he/she doesn't know well." The internal consistency of this scale was good (Cronbach's $\alpha=0.89$).

Parental Social Anxiety

Parents' self-reported social anxiety was measured with the 18-item Social Phobia and Anxiety Inventory (SPAI-18; de Vente et al., 2014). Items are rated on a 7-point Likert scale (ranging from 1 = "never" to 7 = "always") and the social anxiety score is calculated by calculating the sum scores of the 18 items and subtracting 18 from this score. Higher scores indicate higher parental social anxiety. An example item is: "I feel so anxious about attending social gatherings that I avoid these situations." The internal consistency of this scale in the present study was excellent (Cronbach's $\alpha=0.96$).

Fear of Negative Evaluation

Parents' fear of negative evaluation was measured with the Brief Fear of Negative Evaluation Scale (BFNE; Leary, 1983), which is the 12-item version of the Fear of Negative Evaluation Scale (FNE; Watson & Friend, 1969). The correlation between the original and the brief versions is reported to be very high ($r=.96$; Leary, 1983). Parents were asked to rate how much each item applied to them on a 5-point Likert scale (ranging from 1 = Not at all characteristic of me to 5 = Extremely characteristic of me). Parental FNE scores were calculated by taking the sum scores of questionnaire items, with higher scores indicating higher levels of fear of negative evaluation. An example item is: "I am afraid that others will not approve of me." The internal consistency of this scale was excellent (Cronbach's $\alpha=0.93$).

Fear of Negative Child Evaluation

Parents' fear of negative child evaluation was measured with the Fear of Negative Child Evaluation Questionnaire (FNCE-Q; Majdandžić et al., 2008), which consists of 10 items. Participants rate how much each item applies to them on a 5-point Likert scale (ranging from 0 = "not at all characteristic of me" to 4 = "extremely characteristic of me"). The FNCE score is calculated by taking the sum score of questionnaire items, with higher scores indicating higher

parental fear of negative child evaluation. An example item is: "I worry about what kind of impression my child makes on others." The internal consistency of this scale in the current sample was excellent (Cronbach's $\alpha=0.91$).

Interpretation Bias: Scenario Completion Task

The scenario completion task (Mathews & Mackintosh, 2000) is an *online* measure to measure interpretation bias using reaction times (RTs). Self-referent and child-referent interpretation bias scenarios were used to assess self-referent and child-referent interpretation biases, respectively. Participants were asked to read ambiguous scenarios shown on the computer screen. These scenarios described imaginary, but common, social situations in which the parent, or their child was evaluated, and which would potentially trigger social threat concerns. Scenarios about the self-referent interpretation bias were developed by Mathews and Mackintosh (2000). Child-referent scenarios, inspired by the scenarios used by Lester et al. (2009), were developed by the authors. These scenarios concerned social situations in which a child was evaluated. Similar (but age-appropriate) situations were used as in the adult scenarios. Scenarios were presented in 3 lines, with lines following 3500ms after one another. At the end of each scenario, a disambiguating word with a few missing letters was presented 4000ms after the last line of the scenario was shown. This word disambiguated the scenario in either a positive or a negative manner. Parents' task was to press the space bar key as soon as they recognized the missing letters, and to write down these missing letters in a text box on the next screen. The scenario completion trial sequence was interrupted if participants did not press space within 85s. After completing the word fragment, participants were asked to respond to a yes/no comprehension question regarding the content of the scenario. The idea behind the task is that socially anxious participants react faster to the words that resolve the scenarios in a negative manner compared to the positive manner (Mathews & Mackintosh, 2000).

The task included 40 unique scenarios in total, belonging to 4 different categories: 10 positively disambiguated self-referent scenarios, 10 positively disambiguated child-referent scenarios, 10 negatively disambiguated self-referent scenarios, and 10 negatively disambiguated child-referent scenarios. Scenarios were presented in two blocks of 20 scenarios each with a self-paced break in between. Each block contained 5 randomly selected scenarios of each category. An example of a negatively disambiguated child-referent scenario is shown in Fig. 1.

Interpretation bias in this task was operationalized as the difference between the mean RTs for positively versus negatively disambiguated scenarios, separately for self- and

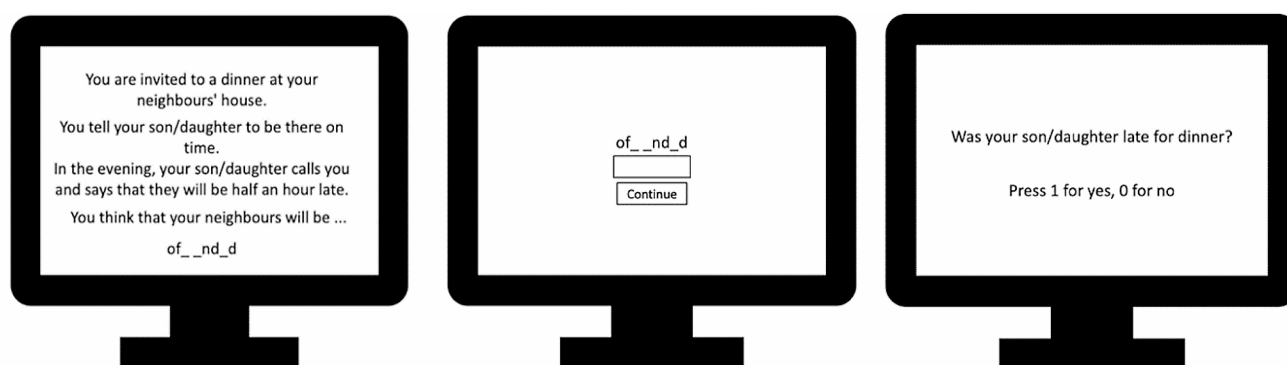


Fig. 1 Schematic example of the trial sequence in the scenario completion task. The disambiguating word for the scenario was “offended”

child-referent scenarios. RTs were measured from the onset of the word fragment until participants pressed the space bar (indicating that they knew how to complete the word fragment). The resulting RT interpretation bias scores indicated the degree of parents’ self-referent and child-referent interpretation bias. A higher score reflects a more negative interpretation bias. Using 5000 random splits, the permutation-based Spearman-Brown corrected split-half reliability of the self-referent interpretation bias scores was $r_{SB} = -0.25$, 95% CI [-0.48, 0.06] and child-referent interpretation bias scores was $r_{SB} = -0.15$, 95% CI [-0.40, 0.16].

Interpretation Bias: Memory Recognition Task

The memory recognition task (Mathews & Mackintosh, 2000) is an *offline* measure to measure interpretation bias. The first part of the memory recognition task was similar to the scenario completion task, with parents completing missing letters of the words in ambiguous social scenarios in which they or their child were evaluated. However, in this task, completing the missing letters of the word did not disambiguate the scenario. Besides, unlike the scenario completion task, each scenario had a title that gave a hint to the content of the scenario to make it easier to remember. Self-referent scenarios were developed by Mathews and Mackintosh (2000). We developed similar child-referent scenarios, based on the contents of the scenario’s used by Lester et al. (2009). An example of an ambiguous child-referent scenario was:

School Manager’s Question

The school manager comes to your son/daughter’s class to monitor the learning process.

The manager asks a question to your son/daughter. Your son/daughter has to think a bit and then comes up with an answer. People in the class are wh_spe_ing” (for whispering)

The task included 10 ambiguous self-referent scenarios and 10 child-referent scenarios. Scenarios were presented in a random order. After completing all scenarios, parents’ memory of the scenarios was tested using both negative and positive endings of the same ambiguous social situations. For the example above, the endings were: “People in the class are whispering about how poor your son/daughter’s answer was” (negative) and “People in the class are whispering about how smart your son/daughter’s answer was” (positive). For each of these endings, parents were required to indicate how similar in meaning the ending was to the scenario that was presented before. The similarity was rated on a 4-point scale (1 = very similar, 2 = fairly similar, 3 = fairly different, 4 = very different). Participants were required to rate the meaning of the sentences independently and were informed that they could choose the same rating for each version. The order of scenarios in this recognition phase was random.

To calculate interpretation bias, we inverted all responses on the similarity scale so that higher scores reflected more agreement (i.e., 1 = very different, 2 = fairly different, 3 = fairly similar, 4 = very similar). Interpretation biases in the memory recognition task were operationalized as the difference between the mean similarity ratings of negative and positive endings (i.e., negative score minus positive score), separately for self-referent and child-referent situations. A higher score on self-referent scenarios reflects parents’ more negative interpretation bias concerning their own environment. Similarly, a higher score on child-referent scenarios reflects parents’ more negative interpretation bias concerning their child’s environment. Using 5000 random splits, the permutation-based Spearman-Brown corrected split half reliability of the self-referent interpretation bias was $r_{SB} = 0.55$, 95% CI = [0.45, 0.64]. For child-referent interpretation bias, the split half reliability was $r_{SB} = 0.62$, 95% CI = [0.53, 0.70]. These reliability estimates are comparable to other studies using this task (Deen et al., 2022; van Bockstaele et al., 2019).

Data Analytical Approach

Correlation coefficients between all study variables were calculated. Before the statistical mediation model analyses, scores were standardized using z -transformation to obtain standardized regression coefficients. Next, to test our hypotheses, we conducted mediation analyses with four different models using the PROCESS macro version 4.1 for SPSS (version 27) developed by Hayes (2022). The PROCESS macro estimates indirect effects through bootstrapping 5000 samples. As there were no significant correlations between parental interpretation biases measured with *online* and *offline* measures, they were tested in different models. First, we tested whether parents' FNE and *online* self-referent interpretation bias parallelly mediated the relation between parent and child social anxiety (Fig. 2). Second, we tested the expected mediation effects of parents' FNE and *offline* self-referent interpretation bias (Fig. 3) in parallel. Third, we examined whether parental FNCE and *online* child-referent interpretation bias parallelly mediated the relation between parent and child social anxiety (Fig. 4). Finally, we tested the expected mediation effects of parental FNCE and *offline* child-referent interpretation bias (Fig. 5) in parallel.

Results

Scoring and Outlier Analysis

From 182 participants who completed the study, 3 participants were excluded from all analyses because they indicated they did not complete the tasks and/or questionnaires to the best of their ability. Thus, our final sample consisted of 179 parents ($M_{\text{age}} = 43.28$ years, $SD = 8.13$).

To ensure the data quality and fidelity further, for the scenario completion task, 44 (0.41%) responses to scenarios in which participants did not respond before the response deadline (85s) were removed. Scenario completions were examined to detect typographic errors (e.g., misspellings, typing a wrong letter closely located to the correct one on the keyboard) and they were counted as correct. Misspellings resulting in different existing English words were coded as errors. Participants whose percentage of correct responses on comprehension questions and word completions deviated more than 3 SD s from the group mean were removed ($M = 0.90$, $SD = 0.08$, $n_{\text{removed}} = 4$ for comprehension questions, and $M = 0.96$, $SD = 0.07$, $n_{\text{removed}} = 2$ for word completions, respectively), as these scores indicate insufficient command of the English language and/or lack of task commitment. For the remaining 173 participants, we removed trials with incorrect word completions (3.30%),

as well as trials with RTs deviating more than 3 SD s from the group mean (2.29%) and trials with RTs deviating more than 3 SD s from participants' individual mean (2.32%). Subsequently, we excluded the participants who had less than 50% of responses left on either positive or negative trials, which indicated that they performed poorly on the corresponding task. However, their data on other tasks were kept if they performed well. For the scenario completion tasks regarding their children, $n = 2$ participants' responses were removed, while $n = 1$ participant's response was removed for the tasks regarding their parents' own environment. For the remaining trials, we calculated the interpretation bias scores based on the differences in RTs in negatively and positively disambiguated scenarios.

For the memory recognition task, we followed a similar procedure. Participants whose percentage of correct responses on comprehension questions and word completions deviated more than 3 SD s from the group mean were removed ($M = 0.89$, $SD = 0.09$, $n_{\text{removed}} = 1$ for comprehension questions, and $M = 0.98$, $SD = 0.04$, $n_{\text{removed}} = 4$ for word completions, respectively). For the remaining 174 participants, interpretation bias scores were then calculated from the mean differences in scenario similarity ratings for negative and positive endings.

Descriptive Statistics

Table 1 shows the descriptive statistics and correlation coefficients for our study's main variables. Parents with higher social anxiety, FNE, FNCE, and *offline* self-referent interpretation bias had children with higher social anxiety. Higher parent social anxiety was associated with higher parent FNE and FNCE. However, parent social anxiety was not significantly associated with parents' *online* and *offline* interpretation biases regarding self-referent and child-referent situations. Parents with higher FNE had higher FNCE and *offline* child-referent interpretation bias. However, parents' FNCE was not significantly associated with parents' *online* and *offline* interpretation biases regarding self-referent and child-referent situations. Parents' interpretation biases based on the *online* measure did not correlate with any of the study variables. As hypothesized, parents' *offline* self- and child-referent interpretation biases were correlated.

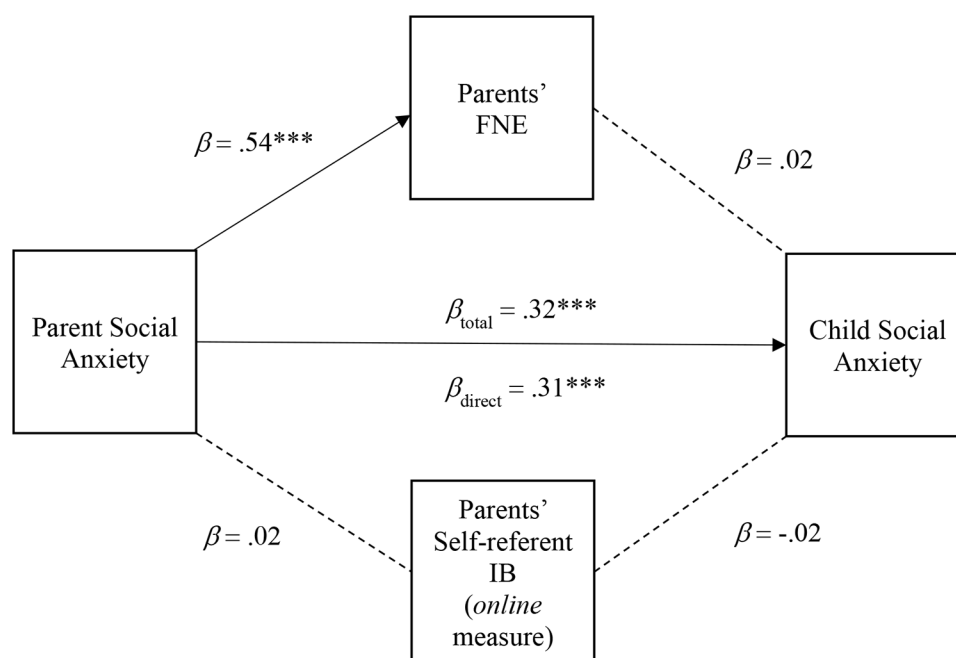
As outcomes of mediation models can be strongly influenced by extreme cases, we investigated multivariate outliers following the guidelines of Aguinis et al. (2013). First, cases exceeding design-specific cut-off points based on the leverage, Cook's distance, Mahalanobis distance, and studentized deleted residuals were identified through 4 separate multiple regression models similar to the hypothesized mediation models. These multiple regression models predicted child social anxiety from parent social anxiety,

Table 1 Descriptive statistics and spearman's correlation coefficients for all study variables

	M (SD)	N	1	2	3	4	5	6	7
1. Child Social Anxiety	7.46 (4.49)	179	-						
2. Parent Social Anxiety	45.97 (23.75)	179	0.34**	-					
3. Parents' FNE	35.06 (12.46)	179	0.20**	0.52**	-				
4. Parents' FNCE	19.99 (9.56)	179	0.27**	0.37**	0.58**	-			
5. Parents' Self-referent IB – Scenario Completion Task	341.32 (1236.49)	172	0.04	0.04	-0.01	-0.07	-		
6. Parents' Child-referent IB – Scenario Completion Task	369.92 (1158.35)	171	-0.14	-0.15*	0.03	0.02	-0.02	-	
7. Parents' Self-referent IB – Memory Recognition Task	-0.25 (0.70)	174	0.18*	0.13	0.11	0.06	0.06	0.08	-
8. Parents' Child-referent IB – Memory Recognition Task	-0.41 (0.70)	174	0.14	0.11	0.21**	0.13	0.07	0.05	0.45**

Note. *** $p < .001$. ** $p < .01$. * $p < .05$. IB = Interpretation bias

Fig. 2 Mediation model of associations between parent and child social anxiety via parents' FNE and self-referent IB - Scenario completion task. Note. *** $p < .001$. ** $p < .01$. * $p < .05$. IB = Interpretation bias



parents' FNE, FNCE, their self-referent and child-referent negative interpretation bias. Then, mediation models were tested with and without these cases. Below, we present the results without the multivariate outliers as our main results.

1

Model 1: Mediation by Parents' FNE and Online Self-referent Interpretation Bias

Model 1 (Fig. 2) tested parents' FNE and *online* self-referent interpretation bias as mediators in the association between parent and child social anxiety. After removing 1 case flagged as potential multivariate outlier ($N=171$), there was a significant total effect of parent social anxiety on child social anxiety. Controlling for the other effects,

parent social anxiety was significantly associated with parents' FNE, but not with parents' *online* self-referent interpretation bias (i.e., measured with the Scenario Completion Task), indicating that parents with high social anxiety had higher FNE but no associations found with self-referent negative interpretation bias based on the *online* measure. Direct effects for parents' FNE ($p = .82$) or their self-referent interpretation bias ($p = .77$) in association with child social anxiety were not statistically significant. Controlling for the effects of parents' FNE and self-referent interpretation bias, parent social anxiety was still significantly associated with child social anxiety. No indirect effects of parent social anxiety on child social anxiety via parents' FNE ($\beta = 0.01$, $SE = 0.05$, 95% CI = $[-0.09, 0.11]$) and *online* self-referent negative interpretation bias ($\beta = -0.00$, $SE = 0.01$, 95% CI = $[-0.01, 0.02]$) were found. Therefore, parents' FNE and their *online* self-referent negative interpretation bias did not

¹ Results based on the entire sample are provided in the online supplement. Results with and without multivariate outliers were largely similar.

significantly mediate the relation between parental social anxiety and child social anxiety.

Model 2: Mediation by Parents' FNE and Offline Self-referent Interpretation Bias

Model 2 (Fig. 3) tested parents' FNE and *offline* self-referent interpretation bias (i.e., measured with the Memory Recognition Task) as mediators in the association between parent and child social anxiety. After removing 1 case flagged as potential multivariate outlier ($N=173$), there was a significant total effect of parent social anxiety on child social anxiety. Holding other variables constant, parent social anxiety had a significant direct effect on parents' FNE but not on parents' *offline* self-referent negative interpretation bias. Controlling for the other variables in the model, only parent social anxiety had a significant direct effect on child social anxiety. No significant indirect effects for parent social anxiety on child social anxiety through parents' FNE ($\beta=0.01$, $SE=0.05$, 95% CI = [-0.09, 0.11]) and their *offline* self-referent negative interpretation bias ($\beta=0.01$, $SE=0.01$, 95% CI = [-0.01, 0.05]) were found, indicating that both parents' FNE and their *offline* self-referent interpretation bias did not significantly mediate the relation between parent social anxiety and child social anxiety.

Model 3: Mediation by Parents' FNCE and Online Child-referent Interpretation Bias

Model 3 (Fig. 4) tested parents' FNCE and *online* child-referent interpretation bias (i.e., measured with the Scenario

Completion Task) as mediators in the association between parent and child social anxiety. After removing 3 cases flagged as potential multivariate outliers ($N=168$), there was again a significant total effect of parent social anxiety on child social anxiety. Holding the variables in the model constant, parent social anxiety was significantly and positively associated with parents' FNCE and not significantly associated with their *online* child-referent negative interpretation bias. Higher parent social anxiety was thus associated with higher FNCE. Neither parents' FNCE ($p=.05$) nor their child referent interpretation bias ($p=.08$) had a significant direct effect on child social anxiety, controlling for the other variables. Controlling for the effects of parents' FNCE and child-referent interpretation bias, the direct effect of parent social anxiety on child social anxiety was still significant, indicating that higher parent social anxiety was still associated with higher child social anxiety. We found a significant indirect effect of parent social anxiety on child social anxiety via parents' FNCE ($\beta=0.06$, $SE=0.03$, 95% CI = [0.002, 0.14]), indicating that parents' FNCE partially mediated the relation between parent and child social anxiety. However, no significant indirect effect was found via parents' child-referent negative interpretation bias ($\beta=0.02$, $SE=0.02$, 95% CI = [-0.004, 0.06]), indicating that this expected partial mediation via interpretation bias was not significant.

Fig. 3 Mediation model of associations between parent and child social anxiety via parents' FNE and self-referent IB - Memory recognition task. Note. *** $p < .001$. ** $p < .01$. * $p < .05$. IB = Interpretation bias

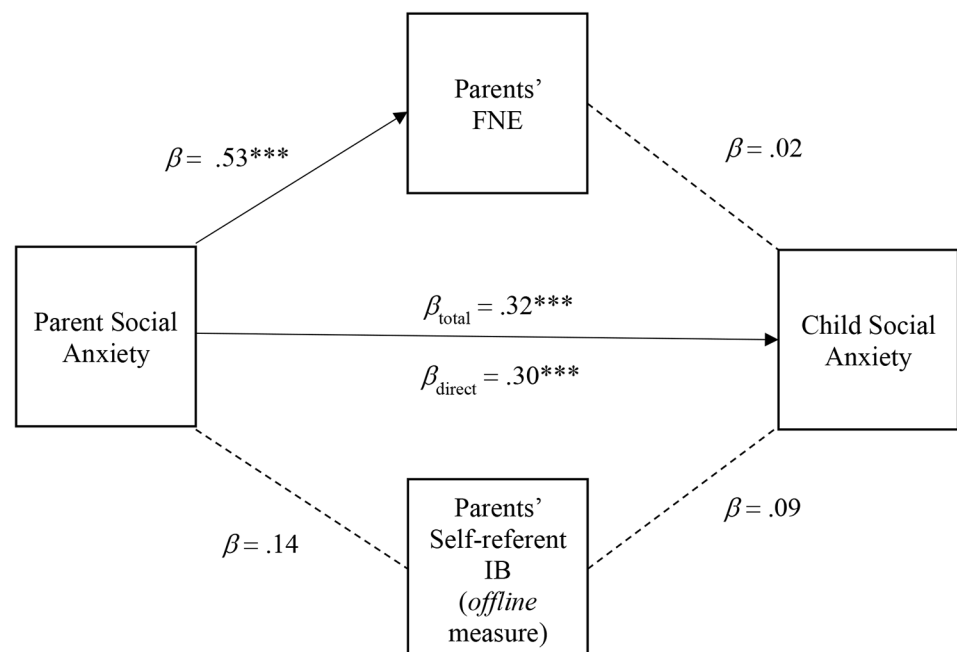


Fig. 4 Mediation model of associations between parent and child social anxiety via parents' FNCE and child-referent IB – Scenario completion task. *Note.* *** $p < .001$. ** $p < .01$. * $p < .05$. IB = Interpretation bias

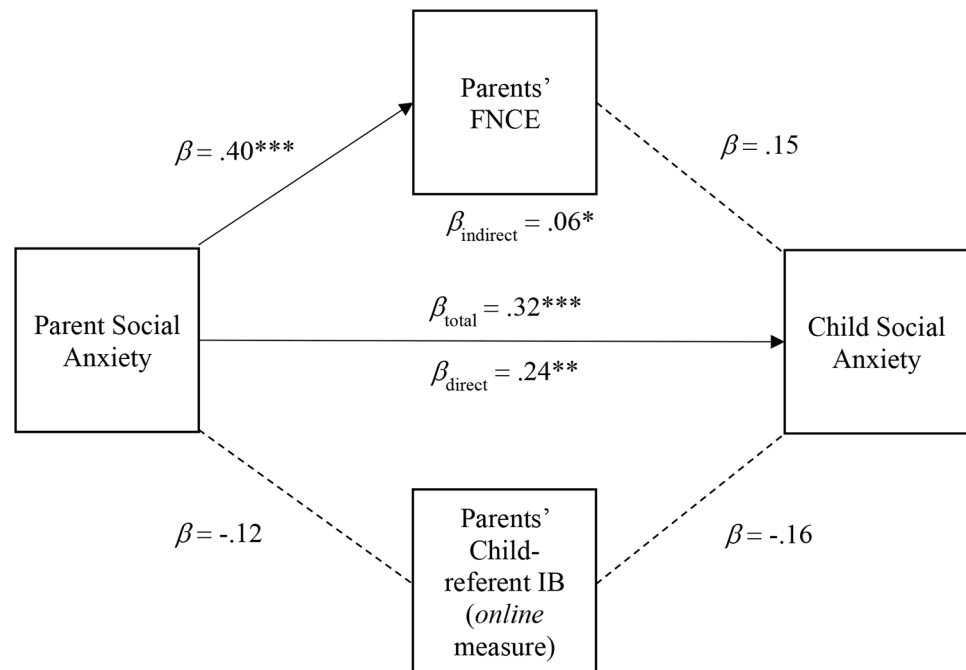
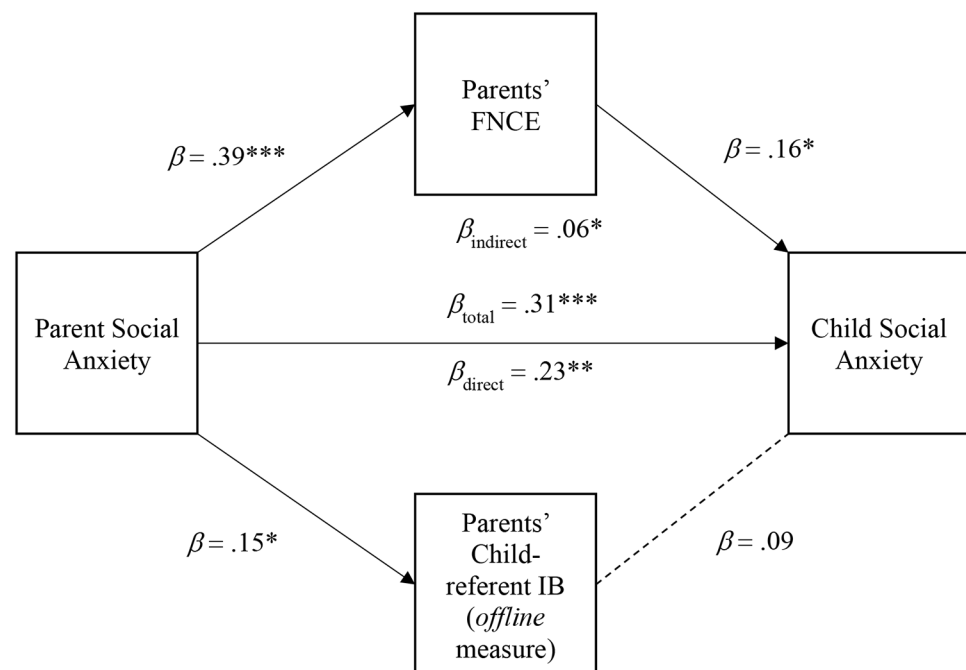


Fig. 5 Mediation model of associations between parent and child social anxiety via parents' FNCE and child-referent IB - Memory recognition task. *Note.* *** $p < .001$. ** $p < .01$. * $p < .05$. IB = Interpretation bias



Model 4: Mediation by Parents' FNCE and Offline Child-referent Interpretation Bias

Model 4 (Fig. 5) tested parents' FNCE and *offline* child-referent interpretation bias (i.e., measured with the Memory Recognition Task) as mediators in the association between parent and child social anxiety. After removing 2 cases flagged as potential multivariate outliers ($N=172$), there was again a significant total effect of parent social anxiety on

child social anxiety. Holding other variables constant, parents with high social anxiety reported significantly higher FNCE, *offline* child-referent negative interpretation bias, and child social anxiety. While parents with higher FNCE reported higher child social anxiety, this direct effect was not found for parents with higher *offline* child-referent negative interpretation bias, holding other variables constant. In addition, there was a significant indirect effect of parent social anxiety on child social anxiety through parents'

FNCE ($\beta = 0.06$, $SE = 0.03$, 95% CI = [0.01, 0.13]), indicating that parents' FNCE partially mediated the association between parent and child social anxiety. However, no significant indirect effect found for parents' *offline* child-referent negative interpretation bias ($\beta = 0.01$, $SE = 0.02$, 95% CI = [-0.01, 0.05]), indicating that the expected partial mediation via interpretation bias was not significant.

Exploratory Analyses

We conducted exploratory analyses to determine whether there were distinctions in the relationships between parent and child social anxiety when fathers' and mothers' FNE and their self-referent interpretation biases, as well as their FNCE and child-referent interpretation were tested in separate models. In Fig. 6a ($n = 71$) and 6b ($n = 100$), we examined whether fathers' and mothers' FNE, along with their *online* self-referent interpretation biases, mediated the link between parental and child social anxiety. Our analysis revealed no mediation effects, and no differences were found between models with mothers versus fathers. In Fig. 6c ($n = 71$) and 6d ($n = 101$), we investigated the

mediation effects of fathers' and mothers' FNE and their *offline* self-referent interpretation biases on the relationship between parental and child social anxiety. Again, our findings did not suggest any significant differences between fathers and mothers in these mediation effects.

As shown in Fig. 7a ($n = 72$) and 7b ($n = 99$), we tested whether fathers' and mothers' FNCE and their *online* child-referent interpretation biases mediated the relation between parent and child social anxiety. While fathers' FNCE significantly mediated the relation between father and child social anxiety ($\beta = 0.11$, $SE = 0.06$, 95% CI = [0.02, 0.23]), this effect was not found in mothers. Additionally, higher social anxiety in fathers was associated with lower *online* child-referent interpretation bias. We also tested whether the relation between parent and child social anxiety was mediated by fathers' and mothers' FNCE and their *offline* child-referent interpretation biases as shown in Fig. 7c ($n = 72$) and 7d ($n = 100$). Similarly, fathers' FNCE significantly mediated the association between father and child social anxiety ($\beta = 0.09$, $SE = 0.05$, 95% CI = [0.01, 0.22]), whereas this effect was not found for mothers. In addition, higher social

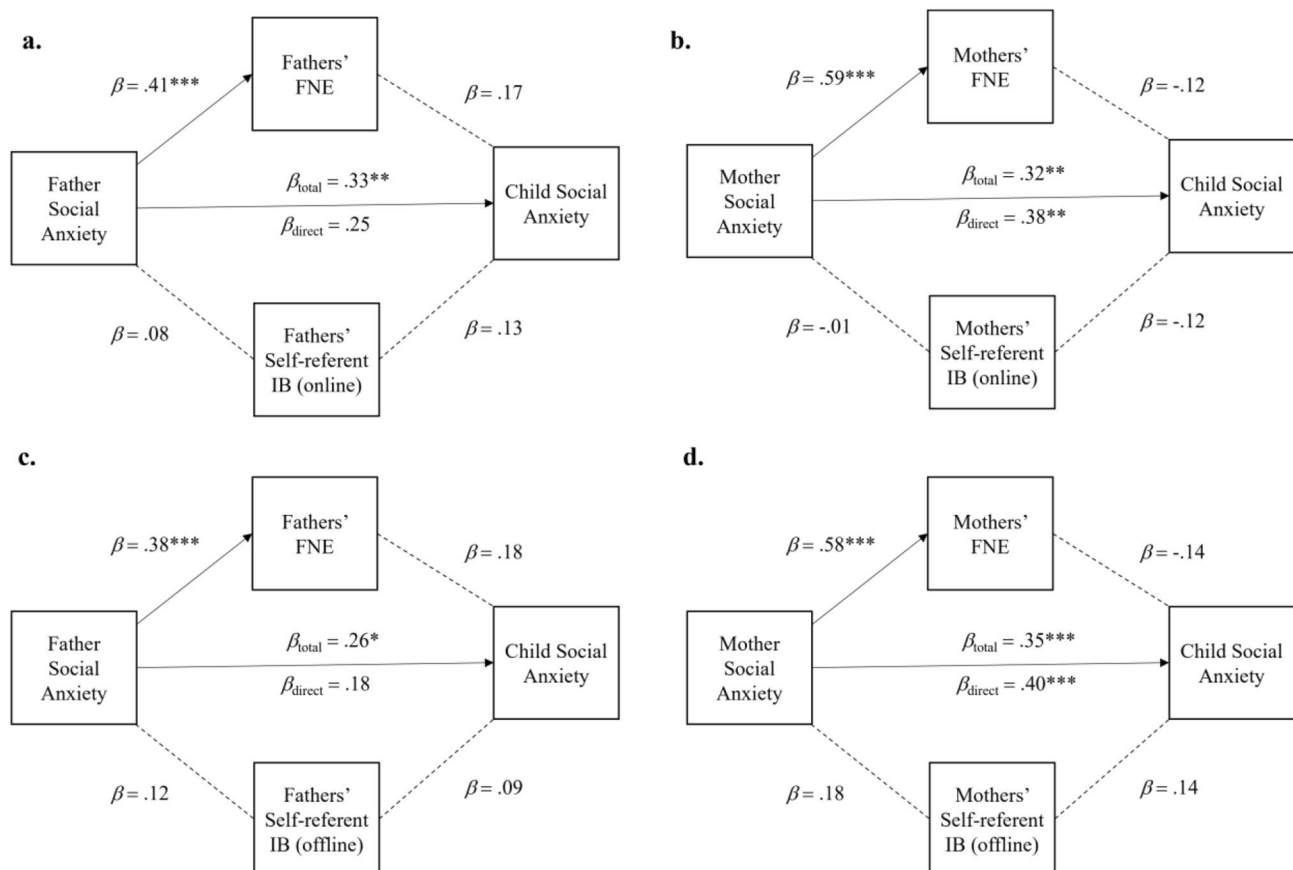


Fig. 6 Models with mediation by mothers' and fathers' FNE and their online and offline self-referent interpretation biases. Note. $***p < .001$. $**p < .01$. $*p < .05$. IB = Interpretation bias

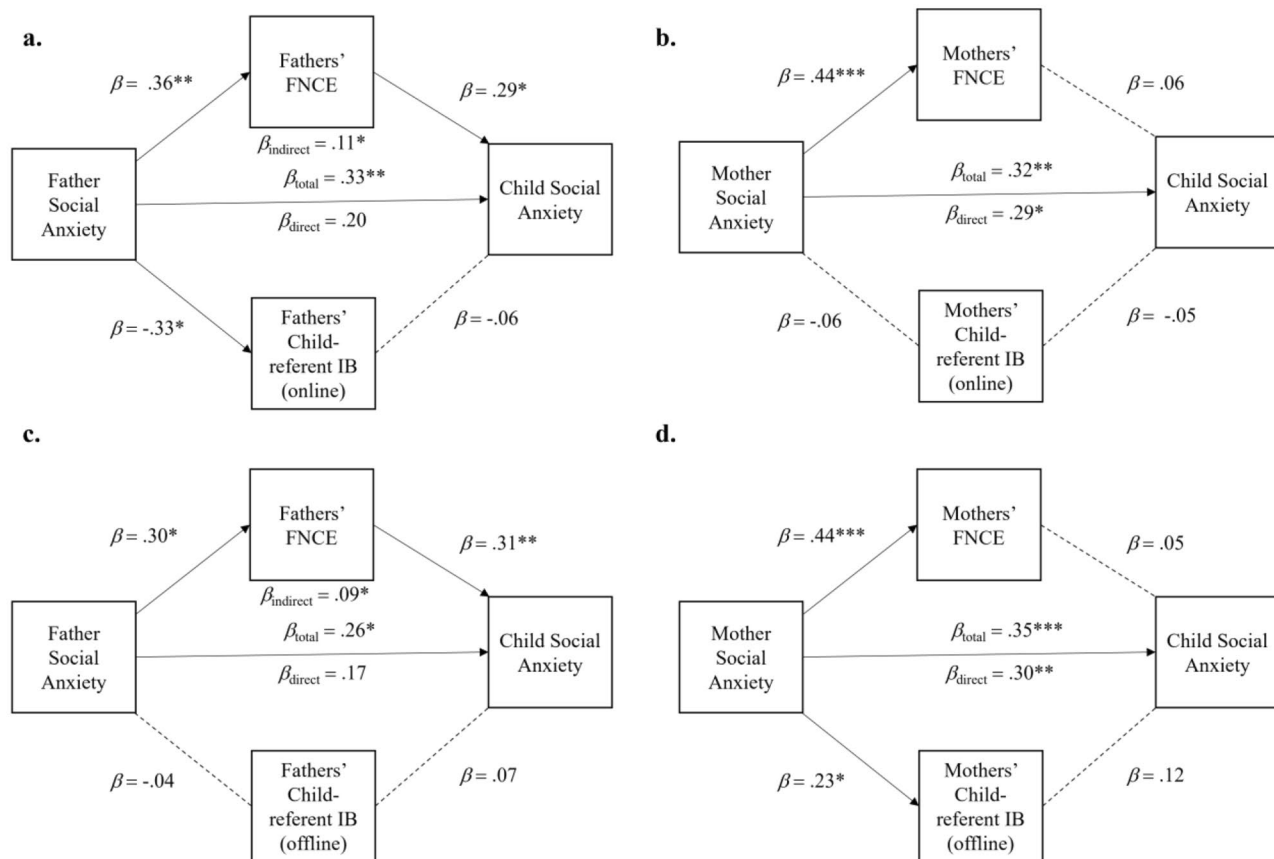


Fig. 7 Models with mediation by mothers' and fathers' FNCE and their online and offline child-referent interpretation biases. *Note.* *** $p < .001$. ** $p < .01$. * $p < .05$. IB = Interpretation bias

anxiety was associated with more *offline* child-referent interpretation bias in mothers.

Discussion

The aim of this study was to investigate whether parents' FNE and self-referent negative interpretation bias, as well as their FNCE and child-referent negative interpretation bias parallelly mediated the relation between parent social anxiety and child social anxiety. We consistently found a relation between parent and child social anxiety, and this relation was statistically partially mediated by parents' FNCE. Parents' FNE and self- and child-referent interpretation biases did not mediate the relation between parent and child social anxiety. In line with our hypotheses, we also found a small to medium, positive, and significant correlation between *offline* self-referent interpretation bias and child social anxiety, but not with parent social anxiety. However, correlations between measures of *online* interpretation bias and indices of both parent and child social anxiety were non-significant. Based on the correlations,

parents with more social anxiety have more concerns about negative evaluation of their child and themselves by others and rate their children as more socially anxious, and parents who interpret ambiguous social situations concerning themselves as more negative rate their child as more socially anxious.

Our finding regarding parents' FNCE as a partial mediator between parent and child social anxiety is in line with the results of three previous studies (de Vente et al., 2011; de Vente et al., 2022; Schreier & Heinrichs, 2010), providing further support for the idea that FNCE plays a significant role in the transmission of social anxiety from parents to children. Children may recognize their parents' anxious reactions and processing of threat information regarding child-referent situations, thus contributing to the aetiology of child social anxiety (Schreier & Heinrichs, 2010). Our results also support the notion that parents' fear of negative evaluation regarding themselves extends to situations involving their children. As such, when parents are anxious about being negatively evaluated by others, they are also worried that their children will be evaluated in that way, be disapproved by others, and make a poor impression on

others (de Vente et al., 2011). While previous studies indicated the role of parents' FNCE in infancy (de Vente et al., 2011) and middle childhood (de Vente et al., 2022), we found that parents' FNCE also plays a role in adolescence (see also Schreier & Heinrichs, 2010). Our exploratory analyses also revealed differential effects for the roles of fathers' and mothers' FNCE. Specifically, we observed an indirect effect of FNCE only for fathers when we tested parental variables in separate models. This suggests a potential unique role of fathers' FNCE in the relationship between parental and child social anxiety. However, these results should be interpreted carefully because of the limited sample size and the exploratory character of these analyses.

Against our hypotheses, our findings suggest that parents' FNE does not mediate the association between parent and child social anxiety. Although de Vente et al. (2022) investigated the extension of parents' FNE to their child's environment (i.e., FNCE), previous studies did not examine parents' FNE as a mediator between parent and child social anxiety. As FNE is related to parents' worries regarding their own environment, it might be less influential in the child's social anxiety context than parents' FNCE, which concerns parents' worries regarding their child's environment. Parents may reflect their child-referent worries more apparently during their parent-child interactions than their self-referent worries.

Our results also indicate that parents' *online* and *offline* self-referent negative interpretation biases do not mediate the relation between parent and child social anxiety. This finding is inconsistent with Podina et al. (2013), who found evidence for a significant partial mediation via parent interpretation bias. However, unlike our study, Podina et al. (2013) investigated the mediation effect of only mothers' self-referent interpretation bias in the association between mother and child social anxiety. Mothers' and fathers' interpretation biases may have different influences, which may lead to diverging results. However, our exploratory analyses in mothers and fathers separately did not reveal any indirect effects for either self-referent or child-referent interpretation biases, assessed through both *online* and *offline* instruments. Thus, unlike Podina et al. (2013), we did not find any mediation effects of mothers' self-referent interpretation bias on the association between parent and child social anxiety. As noted previously, it is important to approach these findings with caution, given the limitations of our sample size.

Our findings also do not correspond with Lester et al. (2009), who found and suggested an extension of parents' interpretation biases to their child's environment. Cognitive models suggest that social anxiety is characterized by interpretation biases (Clark & Wells, 1995), and longitudinal and cognitive bias modification studies provide evidence that adolescents' interpretation biases are causally related

to their social anxiety (Henricks et al., 2022; Krebs et al., 2018). Studies also show different processes such as modeling, social referencing, and direct communication of fears and biases in anxious parents (e.g., Aktar et al., 2017, 2022; Murray et al., 2008), which may cause the transmission of interpretation biases from parents to children. However, our study found no evidence for a relation between social anxiety and self-referent interpretation biases and no evidence for a potential causal mechanism of interpretation biases in the transmission of social anxiety from parent to child. To our knowledge, our study was the first to test the mediation effect via parents' *child*-referent interpretation bias. As such, we cannot compare our findings with other findings in the literature, and future studies need to replicate and extend our findings. Based on our current results however, parents' self-referent and child-referent interpretation biases do not seem to play a role in the intergenerational transmission of social anxiety.

While *online* interpretation bias measures showed only a small significant correlation with parent social anxiety, *offline* interpretation bias measures yielded small significant correlations with child social anxiety and parents' FNE. This finding is consistent with the study of Chen et al. (2020), indicating that the strength of the relation between negative interpretation bias and measures of anxiety is moderated by the type of interpretation bias measure. In our case, we found substantially stronger associations, though not all statistically significant, with parent and child social anxiety for *offline* measures than for *online* measures. Hirsch et al. (2016) argued that *offline* measures of interpretation bias provide respondents time to judge interpretations and allow reflection, which *online* measures do not. As such, compared to *online* measures of interpretation bias, *offline* measures may be more prone to biases other than interpretation bias and are unable to differentiate between inferences made during the first encounter with the stimulus and later inferences (Hirsch et al., 2016). This argument is a possible explanation for our finding that *offline* measures yielded stronger associations with parent and child social anxiety measures than *online* measures. Another possible explanation is that *offline* measures provide time to weigh different interpretations against each other and reveal individuals' reflective processes. Thus, they are better at capturing biased interpretations (Chen et al., 2020).

Unexpectedly, parents' self-referent interpretation biases did not correlate with their social anxiety. This finding is inconsistent with a recent meta-analysis (Chen et al., 2020) showing a strong and positive association between interpretation bias and social anxiety. While Chen et al. (2020) found that the strength of the relation between social anxiety and interpretation bias was dependent on procedural moderators, these moderators cannot explain the absence of

significant relations in our study. Chen et al. (2020) found the strongest effects when interpretation bias was measured using verbal stimuli in *offline* paradigms. In our study, even this relation between *offline* (and verbal) interpretation bias and social anxiety was small and non-significant. The fact that our sample was recruited in the general population, while the meta-analysis was based on clinical and sub-clinical samples with respect to social anxiety disorder may explain these differences in our findings. Another possible explanation for the inconsistent findings could be due to the poor reliability of the interpretation bias tasks, especially for the *online* measure. Previous studies using similar interpretation bias measures have also reported poor reliabilities (e.g., Deen et al., 2022; van Bockstaele et al., 2019). To address the role of interpretation bias in non-clinical samples, future replication studies may consider investigating psychometric properties of the interpretation bias tasks and prefer more reliable measures.

Our study had several limitations. First, all the variables used in the study, including child social anxiety, were based on one and the same parent. Jansen et al. (2017) suggested that there are discrepancies between mother and father reports of child social anxiety. Second, having self-report from one parent may cause informant bias. Socially anxious parents might be biased in their perception of their child's social anxiety (e.g., Cobham et al., 1999) due to their anxious child-referent cognitions. Although a parent's self-perceptions of these constructs were critical, involving the other parent's and/or the child's rating of parental and child social anxiety could provide more accurate estimates of child social anxiety and avoid biased responses due to social desirability. Third, both the sample size and the actual correlations between variables were somewhat lower than we had anticipated to achieve enough power to test the mediations. Nevertheless, our sample size was still large enough to detect meaningful relations between variables. With a sample size of 182, we had enough power (0.80) to detect one-tailed correlations of 0.18 or larger. Additionally, although our sample comprised parents from different countries, around 80% of our participants identified themselves as Caucasian. Previous studies showed that cultural differences influenced the levels of social anxiety (e.g., Jefferies & Ungar, 2020; Schreier et al., 2010), which may affect its transmission. Given the predominantly Caucasian composition of our sample, we acknowledge the limitation in capturing these potential cultural differences. Fourth, as mentioned previously, low reliabilities for the interpretation bias measures might have led to inconsistent results. More reliable measures can be used to replicate the results. Finally, we used a cross-sectional design, so all measures were based on the same time-point, and they were not manipulated. Therefore, we cannot justify causal inferences, nor can we

be certain about the directions of the relationships. It is also important to note that our mediation models refer to the statistical mediation.

Our study provides directions for future studies. First, studies using experimental or longitudinal designs should be conducted to investigate causal relations and justify the directions of the relations between parent and child social anxiety and potential mediators of this relation. Moreover, associations between parent social anxiety, FNCE, child-referent interpretation bias, and child social anxiety may be bidirectional, as higher child social anxiety may also cause higher FNCE and child-referent interpretation biases in parents. Parents of socially anxious children may recognize their child's fear of negative evaluation and interpretation biases, in turn further increasing their FNCE. Future research may manipulate parents' FNE, FNCE, and interpretation biases to examine such possibilities. Additionally, longitudinal studies may be conducted enabling cross-lagged panel modeling to unravel potential bidirectional associations. Second, social anxiety peaks in mid-adolescence (Stein & Stein, 2008). However, younger children may be more vulnerable to parental thoughts and cognitions (Lester et al., 2011), while adolescents may be more susceptible to peer influence (Do et al., 2020). Therefore, future studies may seek to replicate our findings using a younger age group or measure parents' FNCE and interpretation biases in earlier stages of childhood. Finally, mothers and fathers may play differential roles in the intergenerational transmission of social anxiety (de Vente et al., 2011; de Vente et al., 2022; Schreier & Heinrichs, 2010). Although we did not a priori aim to test these differences, our exploratory analyses may serve as evidence for such differences in the role of mothers' and fathers' FNCE. Future studies should explicitly test the differential role of mothers' and fathers' FNCE using larger samples comparing mothers and fathers.

Parents' social anxiety and worries about the social impression of their children may be promising targets for clinical practice. Parents' fear of negative evaluation of their child may be addressed when designing therapies and interventions regarding child social anxiety. However, as causal relations between parents' social anxiety, FNCE, and child social anxiety have not been established yet, further experimental evidence is required before specific clinical implications can be formulated.

In conclusion, we found support for the role of parents' FNCE in the intergenerational transmission of social anxiety in adolescence, providing support for the idea that socially anxious parents are concerned by others' negative evaluations of their children. We found no support for the role of parents' self- and child-referent interpretation biases in the intergenerational transmission of social anxiety during adolescence. In light of our findings, the next important

step is to examine the role of parents' FNCE in the intergenerational transmission of social anxiety using experimental and longitudinal designs.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10608-024-10490-0>.

Funding This study was supported by the Research Institute of Child Development and Education of the University of Amsterdam. Melis Dülger currently works on a doctoral research project at Utrecht University, funded by a grant from the Netherlands Initiative for Educational Research (NRO), with a grant number 40.5.20300.009.

Data availability All data and materials are openly accessible at OSF: <https://osf.io/384h5/>.

Declaration of interest statement The authors declare that they have no conflict of interest.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

- Aguinis, H., Gottfredson, R. K., & Joo, H. (2013). Best-practice recommendations for defining, identifying, and handling outliers. *Organizational Research methods*. SAGE Publications Inc. <https://doi.org/10.1177/1094428112470848>.
- Ahmadzadeh, Y. I., Eley, T. C., Leve, L. D., Shaw, D. S., Natsuaki, M. N., Reiss, D., Neiderhiser, J. M., & McAdams, T. A. (2019). Anxiety in the family: A genetically informed analysis of transactional associations between mother, father, and child anxiety symptoms. *Journal of Child Psychology and Psychiatry and Allied Disciplines*, 60(12), 1269–1277. <https://doi.org/10.1111/jcpp.13068>.
- Aktar, E., Nikolic, M., & Bögels, S. M. (2017). Environmental transmission of generalized anxiety disorder from parents to children: Worries, experiential avoidance, and intolerance of uncertainty. *Dialogues in Clinical Neuroscience*, 19(2), 137–147. <https://doi.org/10.31887/dcns.2017.19.2/eaktar>.
- Aktar, E., Nimphy, C. A., Van Bockstaele, B., & Pérez-Edgar, K. (2022). The social learning of threat and safety in the family: Parent-to-child transmission of social fears via verbal information. *Developmental Psychobiology*, 64(3), e22257. <https://doi.org/10.1002/dev.22257>.
- American Psychiatric Association (2013). *Diagnostic and statistical manual of mental disorders* (5th ed. text revision). <https://doi.org/10.1176/appi.books.9780890425787>.
- Beck, A. T., Emery, G., & Greenberg, R. L. (2005). *Anxiety disorders and phobias: A cognitive perspective*. Basic Books/Hachette Book Group.
- Bögels, S. M., Van Dongen, L., & Muris, P. (2003). Family influences on dysfunctional thinking in anxious children. *Infant and Child Development*, 12(3), 243–252. <https://doi.org/10.1002/icd.288>.
- Bögels, S., Stevens, J., & Majdandžić, M. (2011). Parenting and social anxiety: Fathers' versus mothers' influence on their children's anxiety in ambiguous social situations. *Journal of Child Psychology and Psychiatry and Allied Disciplines*, 52(5), 599–606. <https://doi.org/10.1111/j.1469-7610.2010.02345.x>.
- Birmaher, B., Khetarpal, S., Brent, D., Cully, M., Balach, L., Kaufman, J., & Neer, S. M. K. (1997). The screen for child anxiety related Emotional disorders (SCARED): Scale construction and psychometric characteristics. *Journal of the American Academy of Child and Adolescent Psychiatry*, 36(4), 545–553. <https://doi.org/10.1097/00004583-199704000-00018>.
- Chen, J., Short, M., & Kemps, E. (2020). Interpretation bias in social anxiety: A systematic review and meta-analysis. *Journal of Affective Disorders Elsevier B V*. <https://doi.org/10.1016/j.jad.2020.07.121>.
- Clark, D. M., & Wells, A. (1995). A cognitive model of social phobia. In R. G. Heimberg, M. R. Liebowitz, D. A. Hope, & F. R. Schneier (Eds.), *Social phobia: Diagnosis, assessment, and treatment* (pp. 69–93). The Guilford.
- Cobham, V. E., Dadds, M. R., & Spence, S. H. (1999). Anxious children and their parents: What do they expect? *Journal of Clinical Child and Adolescent Psychology*, 28(2), 220–231. https://doi.org/10.1207/s15374424jccp2802_9.
- Creswell, C., O'Connor, T. G., & Brewin, C. R. (2006). A longitudinal investigation of maternal and child anxious cognitions. *Cognitive Therapy and Research*, 30(2), 135–147. <https://doi.org/10.1007/s10608-006-9021-1>.
- Deen, H., Notebaert, L., Van Bockstaele, B., Clarke, P. J. F., & Todd, J. (2022). For there is nothing either good or bad: A study of the mediating effect of interpretation bias on the association between mindfulness and reduced post-traumatic stress vulnerability. *Bmc Psychiatry*, 22(1). <https://doi.org/10.1186/s12888-022-03950-y>.
- De Lijster, J. M., Dieleman, B. C., Utens, E. M., Dierckx, B., Wierenga, M., Verhulst, F. C., & Legerstee, J. S. (2018). Social and academic functioning in adolescents with anxiety disorders: A systematic review. *Journal of Affective Disorders*, 230, 108–117. <https://doi.org/10.1016/j.jad.2018.01.008>. <https://doi-org.proxy.library.uu.nl/>.
- De Vente, W., Majdandžić, M., & Bögels, S. M. (2022). Intergenerational transmission of social anxiety in childhood through fear of negative child evaluation and parenting. *Cognitive Therapy and Research*. <https://doi.org/10.1007/s10608-022-10320-1>.
- De Vente, W., Majdandžić, M., Colonnese, C., & Bögels, S. M. (2011). Intergenerational transmission of social anxiety: The role of paternal and maternal fear of negative child evaluation and parenting behaviour. *Journal of Experimental Psychopathology*, 2(4), 509–530. <https://doi.org/10.5127/jep.018811>.
- De Vente, W., Majdandžić, M., Voncken, M. J., Beidel, D. C., & Bögels, S. M. (2014). The SPAI-18, a brief version of the social phobia and anxiety inventory: Reliability and validity in clinically referred and non-referred samples. *Journal of Anxiety Disorders*, 28(2), 140–147. <https://doi.org/10.1016/j.janxdis.2013.05.003>.
- Do, K. T., Prinstein, M. J., & Telzer, E. H. (2020). Neurobiological susceptibility to peer influence in adolescence. *The Oxford Handbook of Developmental Cognitive Neuroscience*. Oxford University Press. <https://doi.org/10.1093/oxfordhb/9780198827474.013.27>.
- Eley, T. C., McAdams, T. A., Rijdsdijk, F. V., Lichtenstein, P., Narusyte, J., Reiss, D., Spotts, E. L., Ganiban, J. M., & Neiderhiser, J. M. (2015). The intergenerational transmission of

- anxiety: A children-of-twins study. *The American Journal of Psychiatry*, 172(7), 630–637. <https://doi.org/10.1176/appi.ajp.2015.14070818>.
- Field, A. P. (2006). The behavioral inhibition system and the verbal information pathway to children's fears. *Journal of Abnormal Psychology*, 115(4), 742–752. <https://doi.org/10.1037/0021-843X.115.4.742>.
- Field, A. P., Ball, J. E., Kawycz, N. J., & Moore, H. (2007). Parent-child relationships and the verbal information pathway to fear in children: Two preliminary experiments. *Behavioural and Cognitive Psychotherapy*, 35(4), 473–486. <https://doi.org/10.1017/S1352465807003736>.
- Hayes, A. F. (2022). *Introduction to mediation, moderation, and conditional process analysis, third edition: A regression-based approach*. Guilford Press.
- Heimberg, R. G., Brozovich, F. A., & Rapee, R. M. (2010). A cognitive behavioral model of social anxiety disorder: Update and extension. *Social anxiety* (pp. 395–422). Elsevier Inc. <https://doi.org/10.1016/B978-0-12-375096-9.00015-8>.
- Henricks, L. A., Lange, W. G., Luijten, M., van den Berg, Y. H. M., Stoltz, S. E. M. J., Cillessen, A. H. N., & Becker, E. S. (2022). The longitudinal interplay between attention Bias and Interpretation Bias in Social anxiety in adolescents. *Cognitive Therapy and Research*, 46(5), 940–955. <https://doi.org/10.1007/s10608-022-10304-1>.
- Hirsch, C. R., Meeten, F., Krahé, C., & Reeder, C. (2016). Resolving ambiguity in emotional disorders: The nature and role of interpretation biases. *Annual Review of Clinical Psychology*, 12, 281–305. <https://doi.org/10.1146/annurev-clinpsy-021815-093436>.
- Jansen, M., Bodden, D. H. M., Muris, P., van Doorn, M., & Granic, I. (2017). Measuring anxiety in children: The importance of separate mother and father reports. *Child and Youth Care Forum*, 46(5), 643–659. <https://doi.org/10.1007/s10566-017-9402-5>.
- Jefferies, P., & Ungar, M. (2020). Social anxiety in young people: A prevalence study in seven countries. *PLOS ONE*, 15(9). <https://doi.org/10.1371/journal.pone.023913>.
- Kessler, R. C., Petukhova, M., Sampson, N. A., Zaslavsky, A. M., & Wittchen, H. U. (2012). Twelve-month and lifetime prevalence and lifetime morbid risk of anxiety and mood disorders in the United States. *International Journal of Methods in Psychiatric Research*, 21(3), 169–184. <https://doi.org/10.1002/mpr.1359>.
- Krebs, G., Pile, V., Grant, S., Degli Esposti, M., Montgomery, P., & Lau, J. Y. F. (2018). Research Review: Cognitive bias modification of interpretations in youth and its effect on anxiety: A meta-analysis. *Journal of Child Psychology and Psychiatry and Allied Disciplines*. Blackwell Publishing Ltd. <https://doi.org/10.1111/jcpp.12809>.
- Leary, M. R. (1983). A brief version of the fear of negative evaluation scale. *Personality and Social Psychology Bulletin*, 9(3), 371–375. <https://doi.org/10.1177/0146167283093007>.
- Lester, K., Field, A. P., Oliver, S., & Cartwright-Hatton, S. (2009). Do anxious parents interpretive biases towards threat extend into their child's environment? *Behaviour Research and Therapy*, 47(2), 170–174. <https://doi.org/10.1016/j.brat.2008.11.005>.
- Lester, K. J., Field, A. P., & Muris, P. (2011). Experimental modification of interpretation bias regarding social and animal fear in children. *Journal of Anxiety Disorders*, 25(5), 697–705. <https://doi.org/10.1016/j.janxdis.2011.03.006>.
- Lieb, R., Wittchen, H. U., Höfler, M., Fuetsch, M., Stein, M. B., & Merikangas, K. R. (2000). Parental psychopathology, parenting styles, and the risk of social phobia in offspring. *Archives of General Psychiatry*, 57(9), 859. <https://doi.org/10.1001/archpsyc.57.9.859>.
- Majdandžić, M., de Vente, W., & Bögels, S. M. (2008). *The fear of negative child evaluation questionnaire (FNCE-Q)*. Research Institute Child Development and Education, University of Amsterdam.
- Mathews, A., & Mackintosh, B. (2000). Induced emotional interpretation bias and anxiety. *Journal of Abnormal Psychology*, 109(4), 602–615. <https://doi.org/10.1037/0021-843X.109.4.602>.
- McDowell, D. J., Parke, R. D., & Spitzer, S. (2002). Parent and child cognitive representations of social situations and children's social competence. *Social Development*, 11(4), 469–486. <https://doi.org/10.1111/1467-9507.00211>.
- Murray, L., De Rosnay, M., Pearson, J., Bergeron, C., Schofield, E., Royal-Lawson, M., & Cooper, P. J. (2008). Intergenerational transmission of social anxiety: The role of social referencing processes in infancy. *Child Development*, 79(4), 1049–1064. <https://doi.org/10.1111/j.1467-8624.2008.01175.x>.
- Podina, I. R., Mogoase, C., & Dobrea, A. (2013). Intergenerational transmission of anxiety: Evidence for the mediating role of the negative interpretation bias. *Journal of Cognitive and Behavioral Psychotherapies*, 13(2), 309–320.
- Roefs, A., Huijding, J., Smulders, F. T. Y., MacLeod, C. M., de Jong, P. J., Wiers, R. W., & Jansen, A. T. M. (2011). Implicit measures of association in psychopathology research. *Psychological Bulletin*, 137(1), 149–193.
- Scaini, S., Belotti, R., & Ogliari, A. (2014). Genetic and environmental contributions to social anxiety across different ages: A meta-analytic approach to twin data. *Journal of Anxiety Disorders*, 28(7), 650–656. <https://doi.org/10.1016/j.janxdis.2014.07.002>.
- Schoemann, A. M., Boulton, A. J., & Short, S. D. (2017). Determining power and sample size for simple and complex mediation models. *Social Psychological and Personality Science*, 8(4), 379–386. <https://doi.org/10.1177/1948550617715068>.
- Schreier, S. S., & Heinrichs, N. (2010). Parental fear of negative child evaluation in child social anxiety. *Behaviour Research and Therapy*, 48(12), 1186–1193. <https://doi.org/10.1016/j.brat.2010.09.001>.
- Schreier, S. S., Heinrichs, N., Alden, L., Rapee, R. M., Hofmann, S. G., Chen, J., Oh, K., & Bögels, S. (2010). Social anxiety and social norms in individualistic and collectivistic countries. *Depression and Anxiety*, 27(12), 1128–1134. <https://doi.org/10.1002/da.20746>.
- Stein, M. B., & Stein, D. J. (2008). Social anxiety disorder. *The Lancet*, 371(9618), 1115–1125. [https://doi.org/10.1016/S0140-6736\(08\)60488-2](https://doi.org/10.1016/S0140-6736(08)60488-2).
- Strack, F., & Deutsch, R. (2004). Reflective and impulsive determinants of social behavior. *Personality and Social Psychology Review*, 8(3), 220–247. https://doi.org/10.1207/s15327957pspr0803_1.
- Subar, A. R., & Rozenman, M. (2021). Like parent, like child: Is parent interpretation bias associated with their child's interpretation bias and anxiety? A systematic review and meta-analysis. *Journal of Affective Disorders*, 291, 307–314. <https://doi.org/10.1016/j.jad.2021.05.020>.
- Teachman, B. A., & Allen, J. P. (2007). Development of social anxiety: Social interaction predictors of implicit and explicit fear of negative evaluation. *Journal of Abnormal Child Psychology*, 35, 63–78.
- Telman, L. G. E., van Steensel, F. J. A., Maric, M., & Bögels, S. M. (2018). What are the odds of anxiety disorders running in families? A family study of anxiety disorders in mothers, fathers, and siblings of children with anxiety disorders. *European Child and Adolescent Psychiatry*, 27(5), 615–624. <https://doi.org/10.1007/s00787-017-1076-x>.
- Van Bockstaele, B., Notebaert, L., Salemink, E., Clarke, P. J. F., MacLeod, C., Wiers, R. W., & Bögels, S. M. (2019). Effects of interpretation bias modification on unregulated and regulated emotional reactivity. *Journal of Behavior Therapy and Experimental Psychiatry*, 64, 123–132. <https://doi.org/10.1016/j.jbtep.2019.03.009>.

- Van Bockstaele, B., Tibboel, H., Larsen, H., Wiers, R. W., Bögels, S. M., & Salemink, E. (2021). Dual processes in fear and anxiety: no effects of cognitive load on the predictive value of implicit measures. *Cognition and Emotion*, 35(5), 859–873. <https://doi.org/10.1080/02699931.2021.1898935>
- Watson, D., & Friend, R. (1969). Measurement of social-evaluative anxiety. *Journal of Consulting and Clinical Psychology*, 33(4), 448–457. <https://doi.org/10.1037/h0027806>.
- Williams, J. M. G., Watts, F. N., & MacLeod, C. M. (1997). *Cognitive psychology and emotional disorders* (2nd ed.). Wiley.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.